

Day 3 Review + Gimkit (maybe print these for them)

Today we will be practicing on everything we have learned so far! So let's REVIEW

Find INVERSE

$$22. f(x) = \frac{4x - 1}{2x + 3}$$

NOTATION DECODER

PARENTHESES VS BRACKETS

() means endpoints **NOT** included
 [] means endpoints **included**

Examples:

$(2, 5)$ means $2 < x < 5$

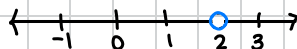
$[2, 5]$ means $2 \leq x \leq 5$

$(2, 5]$ means $2 < x \leq 5$

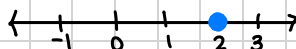
$[2, 5)$ means $2 \leq x < 5$

OPEN VS CLOSED CIRCLE

open circle = **not** included



closed circle (filled) = **included**



INEQUALITIES, INTERVALS AND GRAPHS

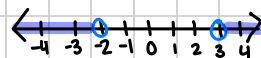
Inequality	Interval Notation	Graph
$x < 3$	$(-\infty, 3)$	
$x \leq 3$	$(-\infty, 3]$	
$x > 3$	$(3, \infty)$	
$x \geq 3$	$[3, \infty)$	

Union Symbol

$\cup$ means union = combine intervals = OR

Ex: $(-\infty, 2) \cup (3, \infty)$

means $x < 2$ or $x > 3$



DOMAIN AND RANGE

Given	Domain	Range	What to watch for								
GRAPH 	all x values the graph touches	all y values the graph touches	watch for arrows, holes, open/closed dots								
Table <table><tr><td>x</td><td>-2</td><td>0</td><td>1/3</td></tr><tr><td>y</td><td>3</td><td>1</td><td>1/7</td></tr></table>	x	-2	0	1/3	y	3	1	1/7	input values (left or top col.)	output values	repeated values count once
x	-2	0	1/3								
y	3	1	1/7								
Formula $f(x) = \frac{1}{x-2}$	<u>allowed</u> x values ↳ consider where restrictions are	possible y values ↳ consider the values of restrictions	watch for denomin. = 0, roots ≥ 0 , ln > 0 , etc.								

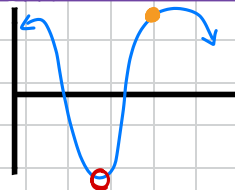
• **Domain** look left → right

• **Range** look down ↑ up

• Arrow continues forever = ∞

• Open dot = endpoint not included

• Filled dot = endpoint included



INVERSE FUNCTIONS

TABLE: SWITCH IN PUT & OUTPUTS

x	-2	0	3
y	1	4	7

x	1	4	7
y	-2	0	3

ordered pairs:

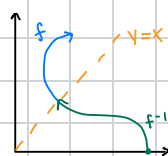
$(a, b) \rightarrow (b, a)$

$(-2, 5) \rightarrow (5, 2)$

$(1, -3) \rightarrow (-3, 1)$

$(4, 7) \rightarrow (7, 4)$

Graph: reflect across the line $y=x$



FORMULA:

- Write $y = f(x)$ replace f(x) w/ y
- switch x & y
- Solve for x

Domain of f^{-1} = Range of f

Range of f = Domain of f^{-1}

⚠ The inverse is a function **ONLY IF** original function is one to one.

DOMAIN & RANGE SWAP

If Dom (f) = $[-3, 5]$ and Range (f) = $[-2, 7]$ then

Dom (f^{-1}) = $[-2, 7]$

Range (f^{-1}) = $[-3, 5]$

SOLVING INVERSE: FORMULA

$$f(x) = 3x - 5$$

- $y = 3x - 5$
- $x = 3y - 5$
- $y = \frac{1}{3}x + \frac{5}{3}$

$f^{-1}(x) = \frac{1}{3}x + \frac{5}{3}$

ONE TO ONE:

each output has only 1 in put.

↳ original function passes HORIZONTAL LINE TEST

